

New Boxes

Monitoring

Deployment orchestration

Deployment orchestration

User Pools

API Gateway

Load Balancer

DNS

Web Backend Layer
N1 N2 N3

Package Infrastructure

Application Layer
N1 N2 N3

Search query Submission

Rapid Development

Event Coordination

Central object Storage

Cached Content

Cache

Relational DB

NoSQL DB

Analytical Processing

Data Warehouse

AWS Services
TALK

Dashboarding

IAM

DNS:- Amazon Route 53

- Define DNS configuration
- Where to route traffic from the internet
- Health checks on your endpoints.
- Any type of traffic shaping you want to do.

Load Balancer:- Elastic Load Balance

- AWS has many more options for a LB but this is the most famous one
 - Application Load Balancer
 - Network Load Balancer

Web Backend / Application Backend:- AWS EC2

AWS Lambda

Serverless Computer infrastructure option, you define just functions that are just code snippets in many programming languages, without worrying about any infrastructure as you did in EC2. Just write and click on deploy (scaling and all is done by AWS)

"Pay for invocation"

↓
Rent virtual machines that you pay for by the hours or time you use. Flexible Service allows you to do whatever you want. (Can be hard to setup.)

AWS ECS → Elastic Container Service

For Docker Fans (like me)

→ Also EKS (Elastic Kubernetes Service)
↳ Variation of ECS

→ Manages our containers, which helps us to
setup server, container and integrated
load balancers and auto scaling helps to facilitate
your deployments to these containers & somewhere
in middle of EC2 and Lambda.

ECS (Elastic Container Service)

~~"Not a simplified version of Kubernetes"~~

• It is an AWS ~~api~~ originated, fully managed
container orchestration service.

• Simplicity

• ~~ECS~~ Easily integrates with App^m/N/w LBs & Cloudwatch

ECS uses its scheduler to determine:

① Where a container is run

② No. of copies started

③ How resources are allocated.

Containerizing the
application →

Push container
image to Amazon
Elastic Container
Repository (ECR)

Define the service
to run the image
in ECS.

EKS (Elastic Kubernetes Service)

"Fully managed Kubernetes cluster"

The primary difference b/w ECS and EKS is how they handle services such as networking and support components.

ECS relies on AWS provided ALB, Route 53, etc

EKS handles all these mechanisms internally just as in any other K8S cluster

EKS provides all the K8S services while managing its managed nature, but increases the application complexity in architecture.

EKS → Pods, Deployments etc

ECS → Directly deploy containers.

→ Calico

Fargate

Even with managed services, servers were still in existence, here comes AWS Fargate Pay-as-you-go serverless compute engine, build and run AWS will manage servers without worrying of user.

"Can't be used in highly regulated environments with strict security and compliance requirements."

As you lose access to underlying servers.

- ECS & Fargate only support AWS VPC n/w mode

→ API Gateway → Additional capability with LB to distribute traffic.

- API Throttling
- API level authorization → integrate it with userpde Service (AWS Cognito)
- Model validation
- Basically as SW layer that presents a single entry point for client to access multiple backend services, while simultaneously managing client server interaction.

Control Plane

Data Plane

→ Amazon Cognito :- All auth related thing "unrelated", can be integrated with other application providers auth.

→ AWS Elastic Cache :- Caching services based on key value lookups. No worry about having infra. relatively hands off.

Memcached Based

Redis Based

→ Relational DBs → Amazon Aurora → AWS RDS

inhouse built amazon DB, compatible with both MySQL & PostgreSQL (Administration, modelling, autoscaling, storage autoscaling, compute autoscaling) provides Data API to call RDS DB using REST API.

Allows user to select which DB config you would like to have. (MySQL, Postgres, etc.) Oracle, Cassandra.

↳ more handson

→ more handsoff

↳ Also have a Serverless module as well

→ NoSQL DBs → DynamoDB

Optimized for key value lookups, Fully managed DB Service
worry → Scaling config, autoscaling is handled behind the scenes.

⊙ ↳ Document DB → MongoDB (type)

→ OpenSearch → It is ~~also~~ Elastic Search behind the scenes

→ Packaged infrastructure → "Conditional of all the required services as a single product"
Bundles all the things and abstracts all the complexity.

- Elastic Beanstalk :- "Old Service" Setup any web app with all things together, kind of an orchestration service, ~~used~~ you are still somewhere accessing the infrastructure here.

- App Runner :- Not that old, uses ECS & Fargate in background. Specify containers do not worry about infra.

"Total abstraction of all low level components"

- Amazon Lightsail :- Similar to other cloud vendors like GCP or DigitalOcean where you can select the type of stack that you want to select for your application. Prepackaged pricing models with reasonable price.

- AppSync → GraphQL Based (fully managed GraphQL as a Service)
↳ Hello Frontend guys!!
↳ Can interface with other AWS backend services.

→ Cached Content → Static files

"Amazon CloudFront" → CDN

How do we deploy and monitor these applications ??

→ For Deployment pipeline

① "Code Commit": Store your source code, or integrate it with github

② "Code Build": Take source code from code commit or any other thing like gitlab and then build that up into artifact, also allows us to create and run test in test env..

CI/CD
vs Build

③ "Code - Deploy": Used to deploy the build's source code to our compute infrastructure.

④ "Code Pipeline": - Deployment orchestration service, allows us to define a workflow with different application stages.

→ For monitoring

① AWS CloudWatch : Contains a lot of different features inside it.

↓
monitor
State of
your
application

→ You can view different metrics on CloudWatch for your services

→ Maintain appⁿ logs

CloudWatch insights → Search capability on a very large cloudwatch data using a SQL styled language.

② AWS CloudTrail : Audit trail of the operations that are performed on the services / infra.

Who performed?
What appⁿ?
What doing?

Identity and access Management (IAM)

Create Pally, associate them with ~~data~~ policies

Rapid Development & Infrastructure as Code

→ IaC

- CloudFormation (JSON/YAML)
- CDK (Cloud development kit): Actual code

→ Rapid development

- AWS ~~Amplify~~ Amplify → Toolkit style service that allows you to build and deploy entire applications.
CLI tool, focuses more on functionality not on infra

It has a set of purpose built tools with travel features that lets frontend web and mobile developers quickly and easily build full stack

Toolset Abstraction applications on AWS. Eliminates need for complex backend config by providing automated testing, auth, data storage & CI/CD

auth in box

ADD API → ON CLI and boom

" Works even if you do not know
other AWS services "

- SPAM (Serverless application Model)

Standard Syntax for IaC, framework designed to simplify, building, testing and deploying serverless applications.

SAM makes our lambda function work locally.

Event Coordination Services

① SNS (Simple Notification Service)

Pub-Sub Service, it is. pub-sub. It publishes a notification to a topic and a topic can have many different subscribers.

② SQS (Simple Queuing Service) : This service is responsible for holding messages so that you can process them at a later time, we define queues and these queues can be connected to many different type of services / compute infra.

When you want people to tell about data or data changes you use 'SNS', if you want to be notified when something changes on in someone else's data you use 'SQS'.

③ EventBridge :- Uses Event Buses (Similar to SNS)
Eg -> integrate an event bridge to an EC2 m/c when it gets terminated, and you write rules to specify who to deliver these events to. Also have sub.

* Eventbridge allows you to define schema definitions of your data to be sent. It helps sales to get access to the models that are going to be sent delivered from the eventbridge by event bus.

* 3rd Party Integrations:- Anytime someone places an order on your shopify ~~you~~ that can be directly integrated into an event bridge.

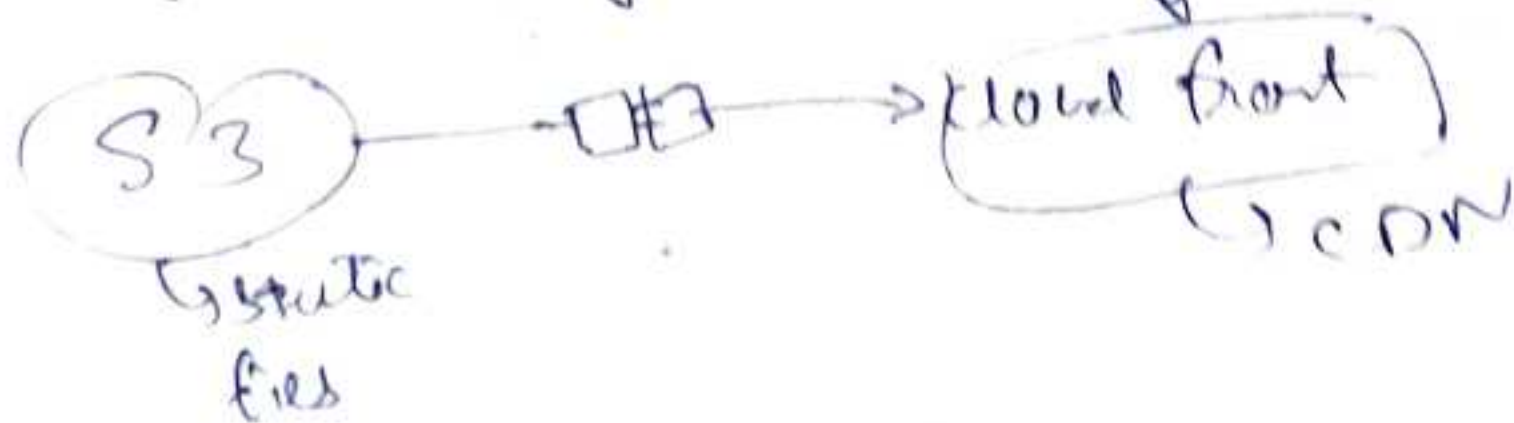
④ Step functions:- Helps you to define workflows
Eg. customer delivery workflows.

Orch

Object Storage → Cold Storage → AWS Glacier

• AWS S3 (Simple Storage Service)

Very cheap, Very scalable, ~~very~~ highly available



Analytical Processing

(S3, R) → Secret
manager
secrets

① EMR → Large Scale distributed data processing system, which allows us to run many different frame works

② Athena → Run queries like SQL
Completely Serverless

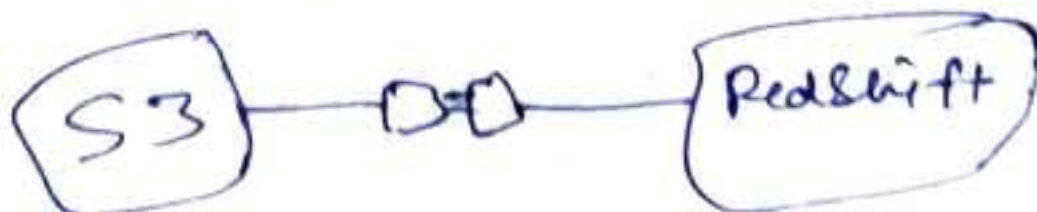
Just store your data in S3 and run
Athena queries on it.
Athena automatically defines your S3
data schema.

Data Warehouse → Central repo used to collect, integrate and store large volumes of current & historical data from multiple disparate sources

① AWS Redshift → Where to store data for business analytics? or any kind of business related stuff?

↓
little costly → It is a columnar style DB that allows us to perform

Also have
a Serverless
mode.



Dashboarding

① QuickSight :- Great for end users, for any data in S3 or Redshift ~~warehouse~~



PowerBI
& AWS

Network boundary

① VPC → Private M/W spaces for you
impr to rest in.

② Subnets

